

chapter 2: Summarizing and Describing the Numerical data

2017-Spring

18) Over a period of 40 days the percentage relative humidity in a vegetable storage building was measured. Mean daily values were recorded as shown below;

60	63	64	71	67	73	79	80	83	81
86	90	96	98	98	99	89	80	77	78
71	79	74	84	85	82	90	78	79	79
78	80	82	83	86	81	80	76	66	74

i) Prepare a stem and leaf display for these data. show the leaves sorted in order of increasing magnitude in each system.

ii) Draw a box plot for these data and interpret the data in practical manner.

Sol Here,
~~Sorting~~

60	63	64	71	67	73	79	80	83	81
86	90	96	98	98	99	89	80	77	78
71	79	74	84	85	82	90	78	79	79
78	80	82	83	86	81	80	76	66	74

i) Stem and leaf display:

Stem | leaves

6	0	3	4	7	7	8	1	9	4	6					
7	1	3	9	7	8	1	9	4	8	9	9	8	6	4	
8	0	3	1	6	9	0	4	5	2	0	2	3	6	1	0
9	0	6	8	8	9	0									

After sorting the leaves:

stem	leaves
6	0 3 4 6 7
7	1 1 3 4 4 6 7 8 8 8 9 9 9 9
8	0 0 0 0 1 1 2 2 3 3 4 5 6 6 9
9	0 0 6 8 8 9

ii) For Box Plot:

$$X_s = 60 \text{ \& } X_L = 99$$

$$n = 40$$

For Q_1 :

$$Q_1 = \left(\frac{n+1}{4} \right)^{\text{th}} \text{ term} = \left(\frac{40+1}{4} \right)^{\text{th}} \text{ term} = 10.25^{\text{th}} \text{ term}$$

$$\text{i.e. } Q_1 = 74 + (76 - 74) \times 0.25 = 74.5$$

For $Q_2 = M_d$:

$$Q_2 = \frac{2(n+1)}{4}^{\text{th}} \text{ term} = 2 \times Q_1 = 2 \times 10.25^{\text{th}} \text{ term} = 20.5^{\text{th}} \text{ term}$$

$$\text{i.e. } Q_2 = M_d = 80 + (80 - 80) \times 0.5 = 80$$

For Q_3 :

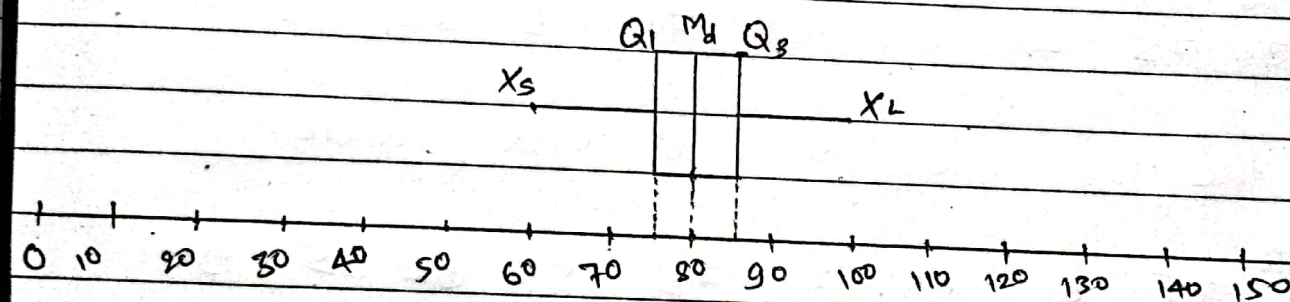
$$Q_3 = \frac{3(n+1)}{4}^{\text{th}} \text{ term} = 3 \times Q_1 = 3 \times 10.25^{\text{th}} \text{ term} = 30.75^{\text{th}} \text{ term}$$

$$\text{i.e. } Q_3 = 84 + (85 - 84) \times 0.75 = 84.75$$

Hence, The five number summary is:
(X_s , Q_1 , M_d , Q_3 , X_L).

i.e. (60 74.5 80 84.75 99)

Now The box plot is given by:



From the graph, the length of box i.e. $Q_3 - Q_1$ is small, which mean there is less variation in data. Also, Q_2 lies approximate middle of Q_1 & Q_3 . so data is semi-symmetrical or partially symmetrical.

2022 - Fall (Pre-Board), NAST

2.6) ~~List~~ The weight (in grams) of 10 orange picked at random from a basket are as follows:

45 30 75 40 60 65 70 85 90 90

i) List the five number summaries.

ii) ~~Form~~ the box and whisker plot ~~represent~~ and describe the shape.

~~sq~~ Here,

45 30 75 40 60 65 70 85 80 90

Arranging the given weights in ascending order,

30 40 45 60 65 70 75 80 ~~90~~ 85 90

$$\therefore X_8 = 30$$

$$\therefore X_L = 90$$

$$\therefore n = 10$$

For Q_1 :

$$Q_1 = \frac{(n+1)}{4}^{\text{th}} \text{ term} = \frac{(10+1)}{4}^{\text{th}} \text{ term} = 2.75^{\text{th}} \text{ term}$$

$$\text{i.e. } Q_1 = 40 + (45 - 40) \times 0.75 = 43.75$$

For $Q_2 = M_d$:

$$Q_2 = \frac{2(n+1)}{4}^{\text{th}} \text{ term} = 2 \times Q_1 = 2 \times 2.75^{\text{th}} \text{ term} = 5.5^{\text{th}} \text{ term}$$

$$\text{i.e. } Q_2 = M_d = 65 + (70 - 65) \times 0.5 = 67.5$$

For Q_3 i.e.

$$Q_3 = \frac{3(n+1)}{4} \text{th term} = 3 \times Q_1 = 3 \times 2.75 = 8.25 \text{th term}$$

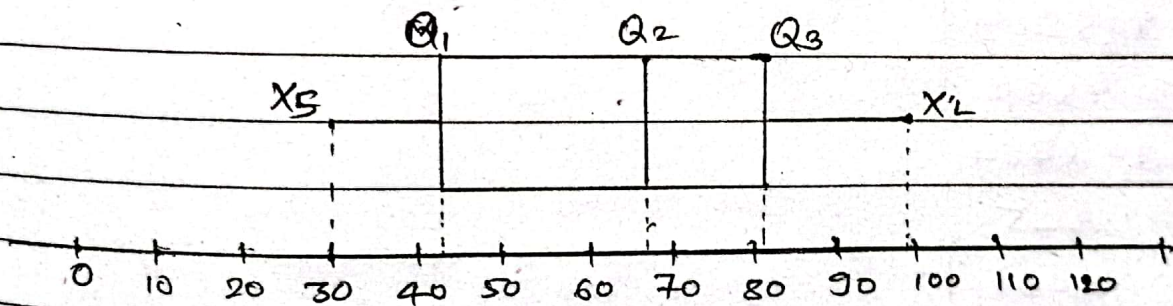
$$\text{i.e. } Q_3 = 80 + (85 - 80) \times 0.25 = 81.25$$

i) The five number summaries:

(x_s Q_1 Q_2 (Md) Q_3 x_L)

i.e. (30 43.75 67.5 81.25 99)

ii) For Box and whisker plot,



From the graph, the length of box i.e. $Q_3 - Q_1$ is large which means there is more variance in data. Also, Q_2 does not lie in the middle of Q_1 and Q_3 , so data is not symmetrical.

2020-Fall

⑤ The nine measurements that follow a furnace temperature recorded on successive batches in the semiconductor manufacturing process (units are);

953, 950, 948, 955, 951, 949, 957, 954, 955

i) Calculate the sample mean, sample variance and standard deviation.

ii) Construct a box plot of the data.

~~Sol~~ Here,

953, 950, 948, 955, 951, 949, 957, 954, 955

$n = 9$

Now,

X	X^2
953	908209
950	902500
948	898704
955	912025
951	904401
949	900601
957	915849
954	910116
955	912025
8572	8164480

i) sample mean, $\bar{x} = \frac{\sum X}{n} = \frac{8572}{9} = 952.44$

$$\begin{aligned}\therefore \text{Sample variance, } \sigma^2 &= \frac{\sum x^2}{n-1} - \left(\frac{\sum x}{n-1} \right)^2 \\ &= \frac{8164430}{9} - \left(\frac{8572}{9} \right)^2 \\ \therefore \sigma^2 &= 8.469 \quad \text{Ans.}\end{aligned}$$

$$\therefore \text{Standard deviation, } \sigma = \sqrt{8.469} = 2.91 \quad \text{Ans.}$$

ii) For box plot:

Arranging in ascending order,

948, 949, 950, 951, 953, 954, 955, 955, 957

$$n=9$$

$$\therefore X_s = 948$$

$$\therefore X_L = 957$$

For Q_1 ,

$$Q_1 = \left(\frac{n+1}{4} \right)^{\text{th}} \text{ term} = \cancel{948} \left(\frac{9+1}{4} \right)^{\text{th}} \text{ term} = 2.5^{\text{th}} \text{ term}$$

$$\therefore Q_1 = 949 + (950 - 949) \times 0.5 = 949.5$$

For $Q_2 = M_d$:

$$Q_2 = \frac{2(n+1)}{4}^{\text{th}} \text{ term} = 2 \times 2.5^{\text{th}} \text{ term} = 5^{\text{th}} \text{ term}$$

$$\therefore Q_2 = 953$$

For Q_3 ,

$$Q_3 = 3(n+1) \frac{th \text{ term}}{4} = 3 \times 2.5 \text{ th term} = 7.5 \text{ th term}$$

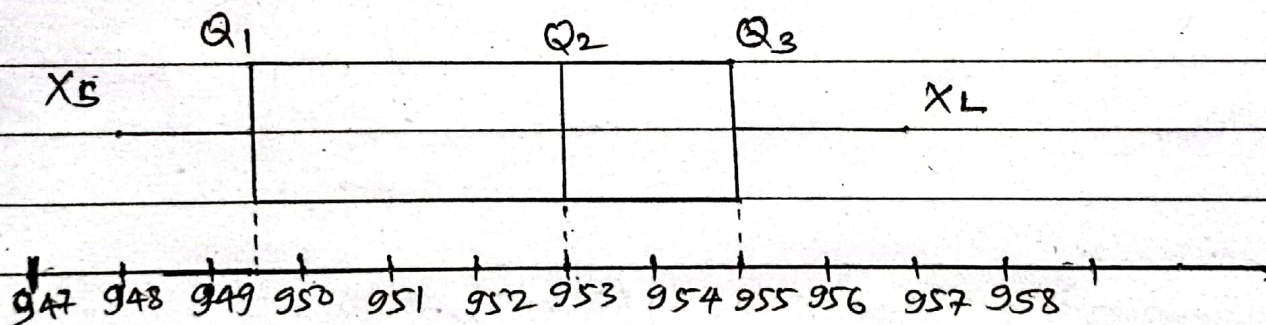
$$\text{i.e. } Q_3 = 955 + (955 - 955) \times 0.5 = 955$$

Five number summary,

(X_s Q_1 Q_2 Q_3 X_L)

i.e. (948 949.5 953 955 957)

For box and whisker plot:



~~From the graph, the length of the box is large~~

2021-Spring

1.6) A random sample was taken of the thickness of insulation in transformer windings, and the following thickness in mm were recorded:

18 21 22 29 25 31 37 38 41 39
44 48 54 56 56 57 47 38 35 36
29 37 32 42 43 40 48 36 37 37

- i) Prepare a stem and leaf display.
- ii) Prepare a box and whisker plot and interpret the shape of the distribution.

eg Here,

18 21 22 29 25 31 37 38 41 39
44 48 54 56 56 57 47 38 35 36
29 37 32 42 43 40 48 36 37 37

i) stem and leaf display:

stem	leaves
1	8
2	1 2 9 5 9
3	1 7 8 9 8 5 6 7 2 6 7 7
4	1 4 8 7 2 3 0 8
5	4 6 6 7

After storing in ascending order,

stem	leaves
1	8
2	1 2 5 9 9
3	1 2 5 6 6 7 7 7 7 8 8 9
4	0 1 2 3 4 7 8 8
5	4 6 6 7

ii) For box and whisker plot;

$$X_S = 18$$

$$X_L = 57$$

$$n = 30$$

For Q_1 ,

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} \text{ term} = \left(\frac{30+1}{4}\right)^{\text{th}} \text{ term} = 7.75^{\text{th}} \text{ term}$$

$$\text{i.e. } Q_1 = 31 + (32 - 31) \times 0.75 = 31.75$$

For Q_2 ,

$$Q_2 = 2\left(\frac{n+1}{4}\right)^{\text{th}} \text{ term} = 2 \times 7.75^{\text{th}} \text{ term} = 15.5^{\text{th}} \text{ term}$$

$$\therefore Q_2 = 37 + (38 - 37) \times 0.5 = 37.5$$

For Q_3 ,

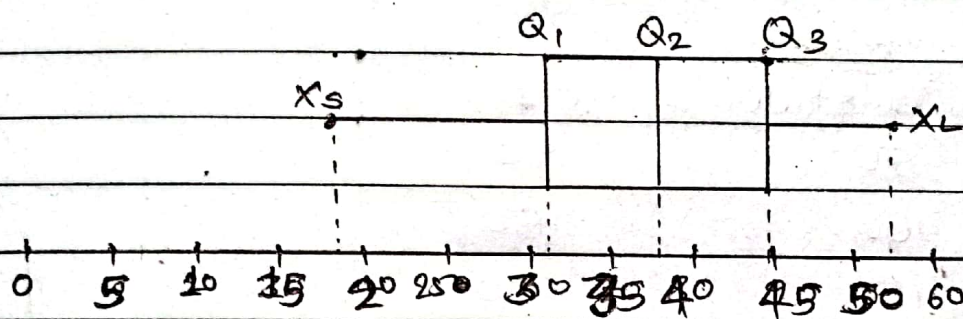
$$Q_3 = 3\left(\frac{n+1}{4}\right)^{\text{th}} \text{ term} = 3 \times 7.75^{\text{th}} \text{ term} = 23.25^{\text{th}} \text{ term}$$

$$\text{i.e. } Q_3 = 44 + (47 - 44) \times 0.25 = 44.75$$

The five number summary is
(x_s , Q_1 , Q_2 , Q_3 , x_L)

i.e (18 31.75 37.5 44.75 57)

Now, The box plot is:



from the graph, the length of the box is large which means there is more variations. And the Q_2 lies in the middle of the box and is symmetrical.

2020=Fall

- (19) From the following distribution of 500 students of a college, find the marks if only 20% students had failed and also minimum marks obtained by the top 25% of the students.

Marks	: 0-20	20-40	40-50	50-60	60-80	80-100
No. of students	: 50	100	150	90	60	50

Also represent the data by histogram to locate the mode.

Here,

Marks	No. of students	c.f.
0-20	50	50
20-40	100	150
40-50	150	300
50-60	90	390
60-80	60	450
80-100	50	500
	N = 500	

If only 20% students had failed, the marks is P_{20} .

For P_{20} ,

$$P_{20} = \left(\frac{20N}{100} \right)^{\text{th}} \text{ term} = \left(\frac{20 \times 500}{100} \right)^{\text{th}} \text{ term} = 100^{\text{th}} \text{ term}$$

$\therefore$ The c.f. just greater than ~~100~~ or equal to 100 is 150. So P_{20} lies in (20-40) interval.

Here,

$$L = 20, \text{ c.f.} = 150, f = 100, h = 20$$

$$\therefore P_{20} = l + \frac{20N/500 - c.f.}{f} \times h = 20 + \frac{100 - 50}{100} \times 20 = 30$$

Hence, the required marks if only 20% students had failed is 30. Ans

Also,

The minimum marks obtained by the top 25% of the students is P_{75} .

For P_{25} ,

$$P_{25} = \left(\frac{75N}{100} \right)^{\text{th}} \text{ term} = \left(\frac{75 \times 500}{100} \right)^{\text{th}} \text{ term} = 375^{\text{th}} \text{ term}$$

$\therefore$ The c.f. just greater or ~~equal~~ than or equal to 375 is 390.80, ~~the~~ P_{25} lies in (50-60) interval.

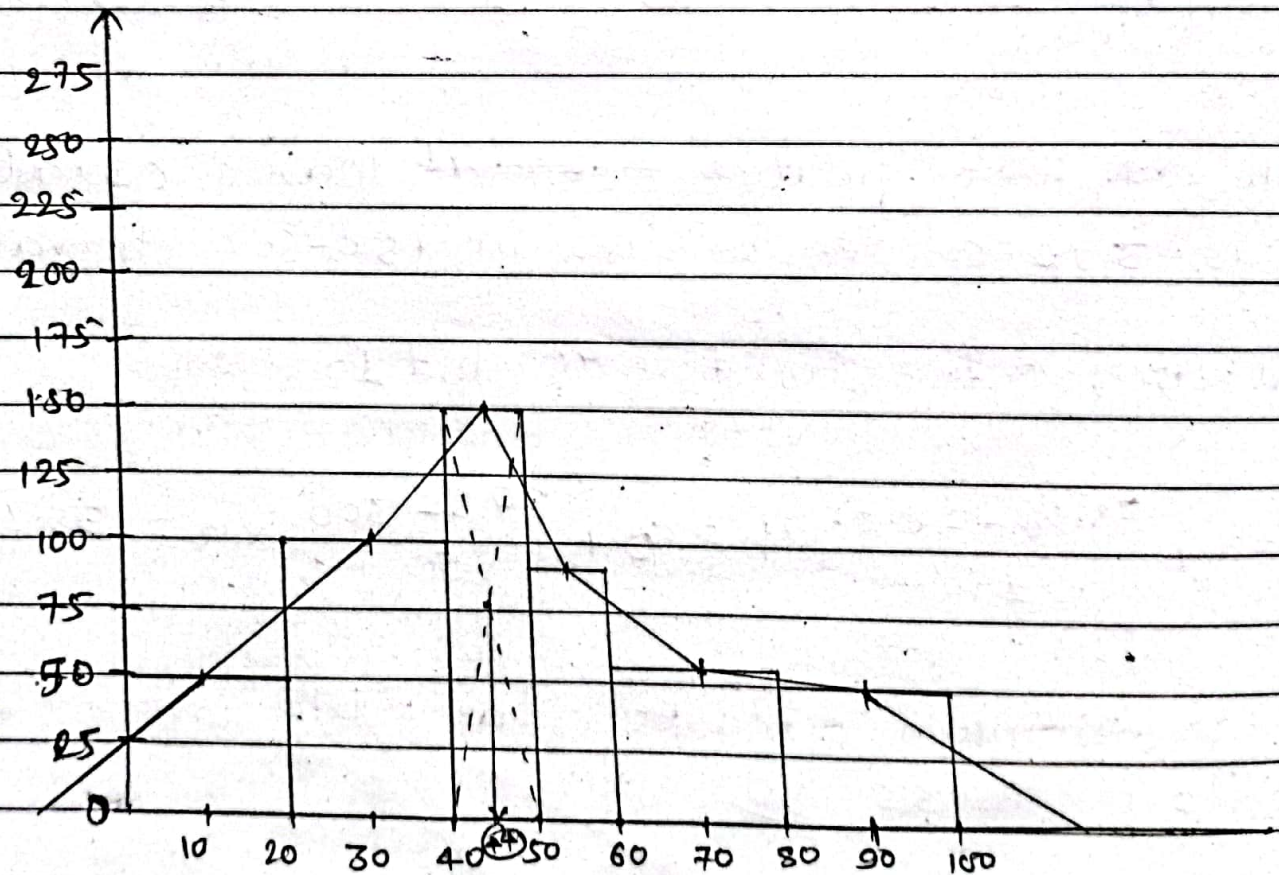
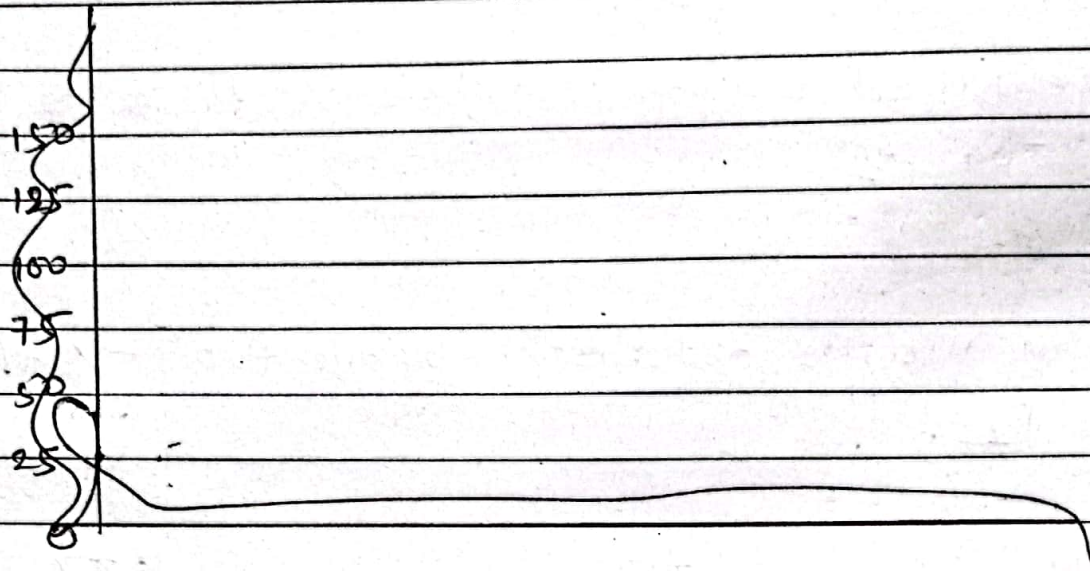
Here,

$$l = 50, c.f. = 300, f = 90, h = 10$$

$$\therefore P_{75} = l + \frac{75N/100 - c.f.}{f} \times h = 50 + \frac{375 - 300}{90} \times 10 = 58.33$$

Hence, the minimum marks obtained by top 25% of the students is 58.33 Ans

Representation of histogram by to locate the mode



Mode = 44

② The table given below represents the daily wages distribution of 130 workers. Find out the range of income of middle 60% workers.

Daily wages (Rs.)	No. of workers
more than 70	130
more than 85	122
more than 100	109
more than 115	79
more than 130	44
more than 145	26
more than 160	14
more than 175	4

Here,

Daily wages (Rs.)	No. of workers (f)	c.f.
70-85	130 8	8
85-100	122 13	21
100-115	109 30	51
115-130	35	86
130-145	18	104
145-160	12	116
160-175	10	126
more than 175	4	130
	N = 130	

The range of income of middle 60% of the workers is ~~Rs. 85-100~~ P₂₀ - P₈₀.

For P_{20} ,

P_{20} lies in $\left(\frac{20N}{100}\right)^{\text{th}}$ term = $\left(\frac{20 \times 130}{100}\right)^{\text{th}}$ term = 26th term.

$\therefore$ The c.f. just greater than or equal to 26 is 51.
So P_{20} lies in the interval (100-115).

Here,

$$\therefore l = 100, f = 30, \text{c.f.} = 21, h = 15$$

$$\therefore P_{20} = l + \frac{20N/100 - \text{c.f.}}{f} \times h = 100 + \frac{26 - 21}{30} \times 15 = \text{Rs. } 102.5$$

For P_{80} ,

P_{80} lies in $\left(\frac{80N}{100}\right)^{\text{th}}$ term = $\left(\frac{80 \times 130}{100}\right)^{\text{th}}$ term = 104th term

$\therefore$ The c.f. just greater than or equal to 104 is 104.80
 P_{80} lies in the interval (130-145).

Here,

$$l = 130, f = 18, \text{c.f.} = 86, h = 15$$

$$\therefore P_{80} = l + \frac{80N/100 - \text{c.f.}}{f} \times h = 130 + \frac{104 - 86}{18} \times 15 = \text{Rs. } 145$$

Hence, the range of the income of middle 60% worker is (Rs. 102.5 - Rs. 145). Ans.

$$\text{i.e. Rs. } 145 - \text{Rs. } 102.5 = \text{Rs. } 42.5 \quad \underline{\text{Ans.}}$$

③ An incomplete frequency distribution is given as follows:

variable	10-20	20-30	30-40	40-50	50-60	60-70	70-80
frequency	12	30	?	65	?	25	18

Given that the median value is 46, determine the missing frequency and total frequency is 229.

Let x and y be the missing frequencies to the corresponding class (30-40) and (50-60) respectively.

C.I.	frequency	c.f.
10-20	12	12
20-30	30	42
30-40	x	$42+x$
40-50	65	$107+x+y$
50-60	y	$107+x+y$
60-70	25	$132+x+y$
70-80	18	$150+x+y$
	$N = 229$	

$$\therefore 150+x+y = 229 \quad \text{--- (1)}$$

Also, median = 46, so, median lies in the interval (40-50).

Here, $l = 40$, $c.f = 42+x$, $f = 65$, $h = 10$

$$\therefore Md = L + \frac{2N/4 - c.f.}{f} \times h$$

$$\Rightarrow 46 = 40 + \frac{2 \times 229/4 - (42 + x)}{65} \times 10$$

$$\Rightarrow \frac{(46 - 40) \times 65}{10} = 114.5 - (42 + x)$$

$$\Rightarrow 39 - 114.5 = -42 - x$$

$$\Rightarrow -75.5 = -42 - x$$

$$\Rightarrow x = -42 + 75.5$$

$$\therefore x = 33.5 \approx 34$$

From (1),

$$150 + x + y = 229$$

$$\Rightarrow 34 + 150 + y = 229$$

$$\Rightarrow y = 229 - 34 - 150$$

$$\therefore y = 45$$

Hence, the missing frequencies are;

$$x = 34 \text{ and } y = 45. \underline{\text{Ans}}$$

④ 100 students took a test, the result of those who secured less than 60% are given below:

Marks	0-20	20-40	40-60
No. of students:	16	24	30

If the average marks of all students was 50, find the average marks of students who secured more than 60% marks.

sol Let $\bar{x}_1$ be the average marks of students who secured less than 60% and $\bar{x}_2$ be the average marks of students who secured more than 60%.

Total students, $n = 100$

Marks(x)	No. of students(f)	Mid-value(x)	fx
0-20	16	10	160
20-40	24	30	720
40-60	30	50	1500
	$n_1 = 70$		$\Sigma fx = 2380$

we know,

$$\bar{x}_1 = \frac{\Sigma fx}{n_1} = \frac{2380}{70} = 34$$

$$\therefore \bar{x}_2 = 50$$

we have, $n_1 + n_2 = n$

$$\Rightarrow 70 + n_2 = 100$$

$$\therefore n_2 = 30$$

Now,

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2} = \frac{70 \times 34 + 30 \bar{X}_2}{100}$$

$$\Rightarrow 50 \times 100 = 70 \times 34 + 30 \bar{X}_2$$

$$\Rightarrow 5000 - 70 \times 34 = 30 \bar{X}_2$$

$$\Rightarrow \bar{X}_2 = \frac{5000 - 70 \times 34}{30}$$

$$\therefore \bar{X}_2 = 87.33$$

Hence, the average marks of those students who secured more than 60% marks is 87.33. Ans

5) From the following data, calculate the percentage of workers getting wages:

i) more than Rs. 44

ii) between Rs. 22 and Rs. 58

wages	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of workers	20	45	85	160	70	55	35	30

Here,

wages	No. of workers:
0-10	20
10-20	45
20-30	85
30-40	160
40-50	70
50-60	55
60-70	35
70-80	30
Σ	$N = 500$

i) The no. of workers with wages more than Rs. 44

$$= \frac{50-44}{10} \times 70 + 55 +$$

$$= \frac{50-44}{10} \times 70 + \frac{60-50}{10} \times 55 + \frac{70-60}{10} \times 35 + \frac{80-70}{10} \times 30$$

$$= Rs. 162$$

$\therefore$ The required percentage is $\frac{162}{500} \times 100\% = 32.4\%$

ii) The no. of workers with wages between Rs. 22 and Rs. 58 is:

$$= \frac{30-22}{10} \times 85 + \frac{40-30}{10} \times 160 + \frac{50-40}{10} \times 70 + \frac{58-50}{10} \times 55$$

$$= \text{Rs. } 342$$

$$\therefore \text{The required percentage} = \frac{342}{500} \times 100\%$$
$$= 68.4\% \quad \underline{\text{Ans}}$$

48 The marks distribution of 104 students is given below:

central rank of group	10	20	30	40	50	60	70
No. of students	7	8	13	29	35	9	3

Find the pass marks of 78 students pass the examinations.

sol Now,

e. Marks	No. of students (f)	c.f.
5-15	7	7
15-25	8	15
25-35	13	28
35-45	29	57
45-55	35	92
55-65	9	101
65-75	3	104
	$N = 104$	

$\therefore$ No. of passed students = 78

$\therefore$ percentage of passed students = $\frac{78}{104} \times 100\%$
 $= 75\%$

Since 75% students are passed that means 25% are failed. so, we need calculate P_{25} in order to get the pass marks of 75% students.

For P_{25} ,

$$P_{25} = \left(\frac{25N}{100} \right)^{\text{th}} \text{ term} = \left(\frac{25 \times 104}{100} \right)^{\text{th}} \text{ ~~student~~ terms}$$
$$= 26^{\text{th}} \text{ term,}$$

Since the c.f. just greater than or equal to 26 is 28, so P_{25} lies in the c.I. (25-35).

Here,

$$l = 25, \text{ c.f.} = 15, f = 13, h = 10$$

$$\therefore P_{25} = l + \frac{25N/100 - \text{c.f.}}{f} \times h$$
$$= 25 + \frac{26 - 15}{13} \times 10$$

$$\therefore P_{25} = 33.46$$

Hence, the pass marks of 78 students is 33.46. Ans

54) From the given data below, find the minimum income earned by the top 15% of the ~~students~~ residents.

Income (00) Rs.	No. of students residents
More than 0	60
more than 100	55
more than 200	40
more than 300	25
more than 400	15
more than 500	10
more than 600	8
more than 700	2

Here,

Income (00) Rs.	No. of residents (f)	c.f.
0 - 100	5	5
100 - 200	15	20
200 - 300	15	35
300 - 400	10	45
400 - 500	5	50
500 - 600	2	52
600 - 700	4	58
700 - 800	2	60
	N = 60	

Since the minimum income earned by the top 15% residents is equal to the maximum income earned by below 85% residents, so, we need to calculate P_{85} to get the minimum income earned by the top 15%.

For P_{85} :

$$P_{85} \text{ lies in } \left(\frac{85N}{100} \right)^{\text{th}} \text{ term} = \left(\frac{85 \times 60}{100} \right)^{\text{th}} \text{ term} \\ = 51^{\text{th}} \text{ term}$$

$\therefore$ The C.f. just greater than ~~50~~ is equal to 51. So the P_{85} lies in the interval (500-600).

Here, $l = 500$, C.f. = 50, $f = 2$, $h = 100$.

$$\therefore P_{85} = l + \frac{\frac{85N}{100} - \text{C.f.}}{f} \times h \\ = 500 + \frac{51 - 50}{2} \times 100$$

$$\therefore P_{85} = 550$$

Hence, the minimum income earned by the top 15% of the residents is Rs. 550 '00'. Ans

(28) One hundred and twenty students appeared for a certain test and the following marks distribution was obtained.

Marks	0-20	20-40	40-60	60-80	80-100
Students	10	30	36	30	14

Find:

- The limit of marks of middle 30% students.
- The percentage of students getting more than 75?
- The number of students who fail if 35 marks are required for passing.

Here,

Marks	Students (f)	c.f.
0-20	10	10
20-40	30	40
40-60	36	76
60-80	30	106
80-100	14	120
	$N = 120$	

Since the limit of marks of middle 30% students is P_{35} and P_{65} .

For P_{35} :

$$P_{35} = \left(\frac{35N}{100} \right)^{\text{th}} \text{ term} = \left(\frac{35 \times 120}{100} \right)^{\text{th}} \text{ term} = 42^{\text{th}} \text{ term}$$

Since the c.f. just greater than or equal to 42 is 76, so P_{35} lies in the interval (40-60).

Here, $l = 40$, c.f. = 40, $f = 36$, $h = 20$

$$\therefore P_{35} = l + \frac{35N/100 - \text{c.f.}}{f} \times h = 40 + \frac{42 - 40}{36} \times 20 \\ = 41.11$$

For P_{85} ,

$$P_{85} = \left(\frac{65N}{100} \right)^{\text{th}} \text{ term} = \left(\frac{65 \times 120}{100} \right)^{\text{th}} \text{ term} = 78^{\text{th}} \text{ term}$$

$\therefore$ The c.f. just greater than or equal to 78 is 106, so P_{85} lies in the interval (60-80).

Here,

$l = 60$, c.f. = 76, $f = 30$, $h = 20$

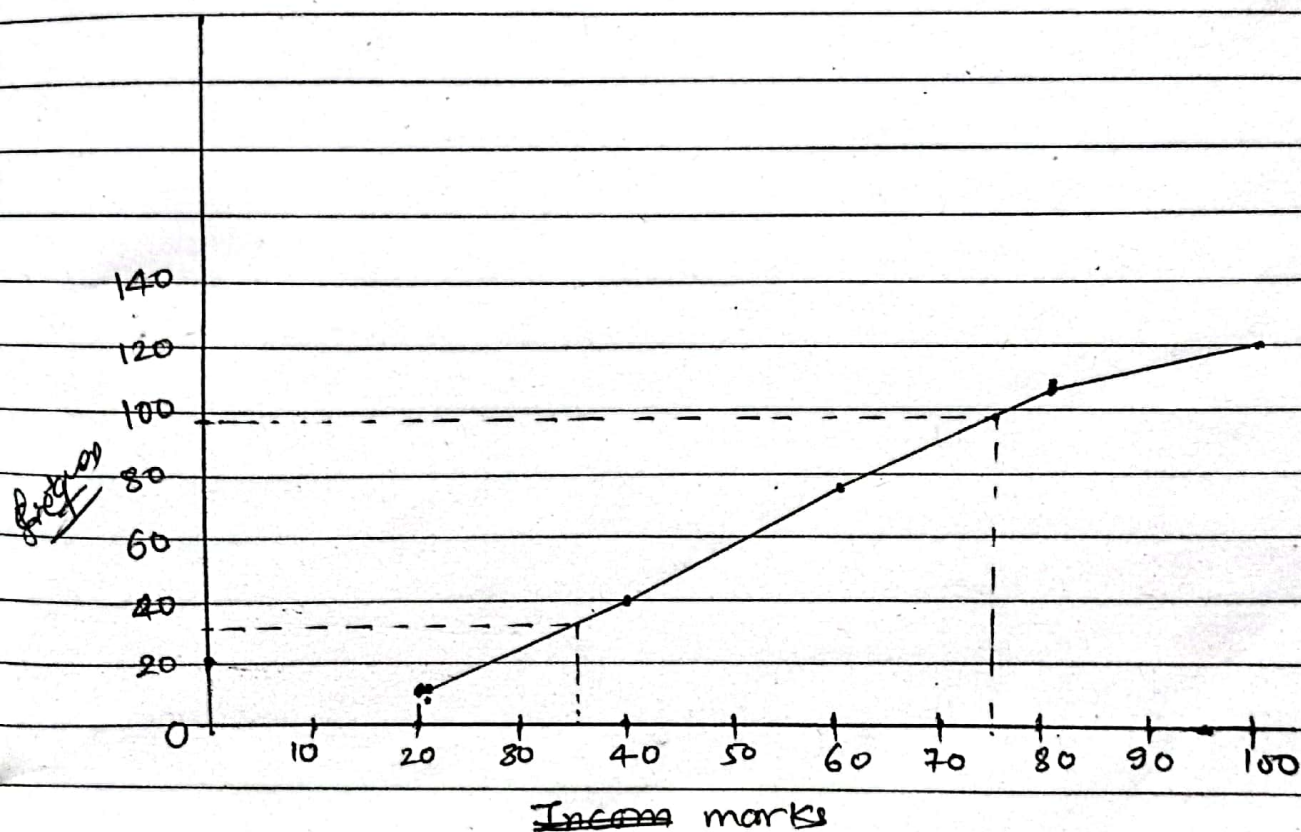
$$\therefore P_{85} = l + \frac{65N/100 - \text{c.f.}}{f} \times h = 60 + \frac{78 - 76}{30} \times 20 \\ = 61.33$$

Hence, the limit of marks of middle 30% students is (41.33 - 61.33) Ans

b>

Now	Marks	C. f
	below 20	10
	below 40	40
	below 60	76
	below 80	106
	below 100	120

The ogive curve is:



b> Hence, the no. of students more than 75 is 98.5

∴ The percentage of students getting more than 75 is $\frac{120 - 98.5}{120} \times 100\% = 17.9\%$

⇒ The number of students who fail if 35 marks are required for passing is 33. Ans

Measures of Dispersion

① A and B have the following scores:

A: 74 101 4 82 36 71 14 0 77

B: 62 5 81 97 22 11 16 1 88

Which score is more uniform?

✍ To check the uniformity of scores, we have to calculate the coefficient of variation of the following scores:

For A scores		For B scores	
x_1	x_1^2	x_2	x_2^2
74	5476	62	3844
101	10201	5	25
4	16	81	6561
82	6724	97	9409
36	1296	22	484
71	5041	11	121
14	196	16	256
0	0	1	1
77	5929	88	7744
459	34879	383	28445

For A scores,

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{459}{9} = 51$$

$$\sigma_1 = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} = \sqrt{\frac{34879}{9} - \left(\frac{459}{9}\right)^2} = 35.70$$

$$\therefore (C.V.)_A = \frac{\sigma_1}{\bar{x}_1} \times 100\% = \frac{35.70}{51} \times 100\% = 70\%$$

For B scores,

$$\bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{383}{9} = 42.56$$

$$\sigma_2 = \sqrt{\frac{\sum x_2^2}{n_2} - \left(\frac{\sum x_2}{n}\right)^2} = \sqrt{\frac{28445}{9} - \left(\frac{383}{9}\right)^2} = 36.74$$

$$\therefore (C.V.)_B = \frac{\sigma_2}{\bar{x}_2} \times 100\% = \frac{36.74}{42.56} \times 100\% = 86.33\%$$

Since $(C.V.)_A < (C.V.)_B$, so, score A is more uniform. Ans

② An analysis of monthly wages paid to the workers in two firms A and B belonging to the same industry gives the following results:

	Firm A	Firm B
No. of workers	500	600
Average monthly wages (Rs.)	186	175
Variance of distribution of wages (Rs.)	81	100

- i) Which firm, A or B has larger wage bill?
- ii) In which firm, A or B, is greater variability in individual wages?
- iii) Calculate the combined mean and combined variance of wage of firm A and firm B.

~~Sol~~

i) For Firm A:

No. of workers, $n_1 = 500$

Average monthly wages, $\bar{x}_1 = 186$

Variance of distribution, $\sigma^2 = 81$

We know that,

$$\text{Average ; } \bar{x}_1 = \frac{\text{Total wages bill}}{\text{No. of workers}}$$

$$\Rightarrow 186 = \frac{\text{Total wages bill}}{500}$$

$$\therefore \text{Total wages bill} = 186 \times 500 = \text{Rs. } 93000$$

For firm B:

No. of workers, $n_2 = 600$

Average wages, $\bar{x}_2 = 175$

Variance of distribution, $\sigma_2^2 = 100$

We know,

$$\text{Average wages, } \bar{x}_2 = \frac{\text{Total wages bill}}{\text{No. of workers}}$$

$$\Rightarrow 175 = \frac{\text{Total wages bill}}{600}$$

$$\Rightarrow \text{Total wages bill} = 175 \times 600$$

$$\therefore \text{Total wages bill} = \text{Rs. } 10,5000$$

Hence, Firm B has ~~more~~ wage bill. Ans
larger

ii) We have,

~~$\sigma_1^2 = 81$~~ for firm A,

$$\sigma_1^2 = 81 ; \bar{x}_1 = 186$$

$$\therefore \sigma_1 = 9$$

$$\therefore (C.V.)_A = \frac{\sigma_1}{\bar{x}_1} \times 100\% = \frac{9}{186} \times 100\% = 4.84\%$$

For firm (B),

$$\sigma_2^2 = 100 ; \bar{x}_2 = 175$$

$$\therefore \sigma_2 = 10$$

$$\therefore (C.V.)_B = \frac{\sigma_2}{\bar{x}_2} \times 100\% = \frac{10}{175} \times 100\% = 5.71\%$$

Since $(C.V.)_A < (C.V.)_B$, so firm B has more variability in individual wages.

iii) We have,

$$\bar{x}_1 = 186 ; n_1 = 500$$

$$\bar{x}_2 = 175 ; n_2 = 600$$

$$\sigma_1^2 = 81 ; \sigma_2^2 = 100$$

$$\therefore \text{combined mean} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{500 \times 186 + 600 \times 175}{500 + 600}$$

$$= \text{Rs. } 180$$

Again,

$$\text{The combined variance, } \sigma^2 = \frac{n_1 (\sigma_1^2 + d_1^2) + n_2 (\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

$$\text{where, } d_1 = \bar{x}_1 - \bar{x} = 186 - 180 = 6$$

$$d_2 = \bar{x}_2 - \bar{x} = 175 - 180 = -5$$

$$\Rightarrow \sigma^2 = \frac{500 (81 + 6^2) + 600 (100 + 25)}{500 + 600}$$

$$\bullet \sigma^2 = 121.36$$

Hence, the combined variance is Rs. 121.36. Ans

③ Lives of two models of refrigerators turned in for new model in a recent survey.

Life (No. of years)	Number of Refrigerators	
	For model A	For model B
0-2	5	2
2-4	16	7
4-6	13	12
6-8	17 7	19
8-10	5	9
10-12	4	1

What is the average life of each model of these refrigerators?
Which model is more uniform?

for model A:

Life (No. of years)	Mid-value (x)	frequency (f)	fx	fx^2
0-2	1	5	5	5
2-4	3	16	48	144
4-6	5	13	65	325
6-8	7	7	49	343
8-10	9	5	45	405
10-12	11	4	44	484
		$N=50$	256	1706

Now,

$$\bar{X}_A = \frac{\sum fx}{N} = \frac{256}{50} = 5.12$$

$$\sigma_A = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} = \sqrt{\frac{1706}{50} - \left(\frac{256}{50}\right)^2} = 2.81$$

$$\therefore (C.V.)_A = \frac{\sigma_A}{\bar{X}_A} \times 100\% = \frac{2.81}{5.12} \times 100\% = 54.88\%$$

For Model B:

Life (No. of years)	Mid-value (x)	frequency (f)	fx	fx ²
0-2	1	2	2	2
2-4	3	7	21	63
4-6	5	12	60	300
6-8	7	19	133	931
8-10	9	9	81	729
10-12	11	1	11	121
		N=50	308	2146

$$\therefore \bar{X}_B = \frac{\sum fx}{N} = \frac{308}{50} = 6.16$$

$$\therefore \sigma_B = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} = \sqrt{\frac{2146}{50} - \left(\frac{308}{50}\right)^2} = 2.23$$

$$\therefore (C.V.)_B = \frac{\sigma_B}{\bar{X}_B} \times 100\% = \frac{2.23}{6.16} \times 100\% = 36.20\%$$

Hence, the average life of each model of these refrigerators is:

$$\bar{X}_A = 5.12 \text{ yrs and } \bar{X}_B = 6.16 \text{ yrs. } \underline{AD}$$

Since $(C.V.)_A > (C.V.)_B$, so the model B has more uniformity.

(2023-Spring)

1. b) The average electricity consumption and standard deviation of two sets of ~~thous~~ households are as follows:

	set I	set II
Mean	65 units	72 units
Standard deviation	13 units	11 18 units
No. of households	50	40

i) Which set is more variable?

ii) Calculate the coefficient of the combined set.

sol

i) For set I:

$$\text{Mean, } \bar{X}_1 = 65$$

$$\text{S.D., } \sigma_1 = 13$$

$$\text{No. of household, } n_1 = 50$$

$$\therefore (C.V.)_I = \frac{\sigma_1}{\bar{x}_1} \times 100\% = \frac{13}{65} \times 100\% = 20\%$$

For set II:

$$\text{Mean, } \bar{x}_2 = 72$$

$$S.D., \sigma_2 = 18$$

$$\text{No. of households, } n_2 = 40$$

$$\therefore (C.V.)_{II} = \frac{\sigma_2}{\bar{x}_2} \times 100\% = \frac{18}{72} \times 100\% = 25\%$$

$\therefore (C.V.)_I < (C.V.)_{II}$. So set II is more variable.

ii) Now,

$$\begin{aligned} \text{Combined mean, } \bar{x} &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{50 \times 65 + 40 \times 72}{80 + 40} \\ &= \frac{5570}{120} = 68.11 \end{aligned}$$

Now,

Combined,

$$\text{standard deviation, } \sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

where,

$$d_1 = \bar{x}_1 - \bar{x} = 65 - 68.11 = -3.11$$

$$d_2 = \bar{x}_2 - \bar{x} = 72 - 68.11 = 3.89$$

$$\Rightarrow \sigma = \sqrt{\frac{50(13^2 + (-3.11)^2) + 40(18^2 + (3.89)^2)}{50 + 40}} = 15.81$$

$$\therefore \text{Combined C.V.} = \frac{S}{\bar{X}} \times 100\% = \frac{15.81}{68.11} \times 100\% = 23.21\%$$

Hence, the coefficient of variation of the combined set is 23.21%. ~~Ans~~

~~2020~~ (2022-Fall)

1. b) The scores of two golfers for 10 rounds each are:

A:	58	59	60	54	65	66	52	75	69	52
B:	84	56	92	65	86	78	44	54	78	68

Which may be regarded as the more consistent player,

~~Here,~~

For A:

X	58	59	60	54	65	66	52	75	69	52	= 610
X ²	3364	3481	3600	2916	4225	4356	2704	5625	4761	2704	= 37736

$$N = 10$$

$$\therefore \bar{X}_A = \frac{\sum X}{N} = \frac{610}{10} = 61$$

$$\therefore \sigma_A = \sqrt{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2} = \sqrt{\frac{37736}{10} - \left(\frac{610}{10}\right)^2} = 7.25$$

$$\therefore (C.V.)_A = \frac{\sigma_A}{\bar{X}_A} = \frac{7.25}{61} \times 100\% = 11.88\%$$

For B:

x_i :	84	56	92	65	86	78	44	54	78	68	$\Sigma = 705$
x_i^2 :	7056	3136	8464	4225	7396	6084	1936	2916	6084	4624	$\Sigma = 51921$

$$\therefore \bar{X}_B = \frac{\Sigma X}{N} = \frac{705}{10} = 70.5$$

$$\therefore \sigma_B = \sqrt{\frac{\Sigma x^2}{N} - \left(\frac{\Sigma X}{N}\right)^2} = \sqrt{\frac{51921}{10} - \left(\frac{705}{10}\right)^2} = 14.89$$

$$\therefore (C.V.)_B = \frac{\sigma_B}{\bar{X}_B} \times 100\% = \frac{14.89}{70.5} \times 100\% = 21.12\%$$

$\therefore (C.V.)_A < (C.V.)_B$. So the player A is regarded as more consistent player.

(2021-Spring)

1. a) The following are the marks obtained by two students A and B in 10 tests 100 marks each:

Test:	1	2	3	4	5	6	7	8	9	10
Marks by A	44	80	76	77	65	72	68	56	90	80
Marks by B	48	75	54	60	63	69	72	51	57	66

Determine

i) who is better student?

ii) Who is more consistent?

Ex 9:

Marks by A:

X:	44	80	76	77	65	72	68	56	90	80	= 708
X ² :	1936	6400	5776	5929	4225	5184	4624	3136	8100	6400	= 51710

Now,

$$\therefore \bar{X}_A = \frac{\sum X}{N} = \frac{708}{10} = 70.8$$

$$\therefore G_A = \sqrt{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2} = \sqrt{\frac{51710}{10} - \left(\frac{708}{10}\right)^2} = 12.58$$

$$\therefore (C.V.)_A = \frac{G_A}{\bar{X}_A} \times 100\% = \frac{12.58}{70.8} \times 100\% = 17.77\%$$

Marks by B:

X:	48	75	54	60	63	69	72	51	57	66	= 615
X ² :	2304	5625	2916	3600	3969	4761	5184	2601	3249	4356	= 38565

Now,

$$\therefore \bar{X}_B = \frac{\sum X}{N} = \frac{615}{10} = 61.5$$

$$\therefore G_B = \sqrt{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2} = \sqrt{\frac{38565}{10} - \left(\frac{615}{10}\right)^2} = 8.62$$

$$\therefore (C.V.)_B = \frac{G_B}{\bar{X}_B} \times 100\% = \frac{8.62}{61.5} \times 100\% = 14.02\%$$

∴ Student A is better student because he secured more marks than B.

ii) B is more consistent because $(C.V.)_A > (C.V.)_B$.

(2021-Fall)

1.6) The life of two model A and B of polythene bags in a recent survey shown below:

Brusting pressure (in lb)	No. of polythene bags	
	A	B
5-10	2	9
10-15	9	11
15-20	29	18
20-25	44	32
25-30	11	17
30-35	5	13

So, Here,

For model A:

Brusting Press.	Mid-value (x)	Frequency (f)	fx	fx ²
5-10 7.5	7.5	2	15	112.5
10-15 12.5	12.5	9	112.5	1406.25
15-20 17.5	17.5	29	507.5	8881.25
20-25 22.5	22.5	44	990	22275
25-30 27.5	27.5	11	302.5	8318.75
30-35 32.5	32.5	5	162.5	5281.25
		N = 100	2090	46275

Also,

$$\bar{x}_A = \frac{\sum fx}{N} = \frac{2090}{100} = 20.9$$

$$\sigma_A = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} = \sqrt{\frac{46275}{100} - \left(\frac{2090}{100}\right)^2} = 5.09$$

$$\therefore (C.V.)_A = \frac{\sigma_A}{\bar{x}_A} \times 100\% = \frac{5.09}{20.9} \times 100\% = 24.35\%$$

For model B:

Brusting pressure	Midvalue (x)	frequency (f)	fx	fx ²
5-10	7.5	9	67.5	506.25
10-15	12.5	11	137.5	1718.75
15-20	17.5	18	315	5512.5
20-25	22.5	32	720	16200
25-30	27.5	17	467.5	12856.25
30-35	32.5	13	422.5	13731.25
		N = 100	2130	50525

Now,

$$\therefore \bar{x}_B = \frac{\sum fx}{N} = \frac{2130}{100} = 21.3$$

$$\therefore \sigma_B = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} = \sqrt{\frac{50525}{100} - \left(\frac{2130}{100}\right)^2} = 7.18$$

$$\therefore (C.V.)_B = \frac{\sigma_B}{\bar{x}_B} \times 100 = \frac{7.18}{21.3} \times 100\% = 33.71\%$$

i) Hence, the average life of model A is 20.9 and the average life of model B is 21.3.

ii) Since $(C.V.)_A < (C.V.)_B$, so model A has greater uniformity because model B has more coefficient of variation than model A.